Land Cover Classification in SAR Time Series Using Ordinal Pattern Features

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***Abstract*—We investigate ordinal-pattern-based features for Land Use/Land Cover (LULC) classification of Synthetic Aperture Radar (SAR) time series data. Temporal dynamics are characterized through normalized Shannon entropy, Shannon statistical complexity, Fisher Information Measure (FIM), and Fisher-based complexity derived from ordinal pattern distributions. The methodology is applied to SAR time series from five LULC classes: Agriculture, Deforested, Forest, Urban, and Water. The resulting features are classified with a Support Vector Machine (SVM). The information-complexity representation reveals partial class separation, with Agriculture exhibiting the clearest distinction from the remaining classes, while substantial overlap persists among Forest, Deforested, Urban, and Water samples. Classification results show that Agriculture and Forest achieved the highest sensitivities (83.3 % and 91.7 %, respectively), whereas Urban and Water remain the most challenging LULC classes to distinguish. These findings indicate that ordinal-pattern-based features capture meaningful temporal characteristics of SAR time series and provide useful information for LULC classification, although discrimination among classes with similar temporal dynamics remains challenging.**

***Index Terms*—Ordinal patterns, SAR time series, land cover classification, Shannon entropy, Fisher Information Measure, statistical complexity, support vector machines.**

I. Introduction

Time series data often show complex patterns over time, and understanding how these patterns are generated is important in many scientific and engineering fields. Traditional methods usually analyse data in either the time domain or the frequency domain, and often assume properties such as linearity, stationarity, or Gaussian behaviour. However, real world data frequently violate these assumptions, creating the need for more flexible, non-parametric approaches.

Recent advances in Synthetic Aperture Radar (SAR) imaging provide dense temporal observations that enable detailed monitoring of land cover dynamics (Sabir et al., 2024). However, SAR time series are often noisy and exhibit complex, nonlinear temporal behaviour, making reliable classification difficult (Sabir et al., 2025). This challenge motivates the use of ordinal-pattern-based methods, which analyse the relative ordering of values rather than their absolute magnitudes. By focusing on temporal structure, these methods are inherently robust to monotonic transformations, speckle-related fluctuations, and outliers, making them well suited for capturing meaningful temporal signatures in SAR data.

Ordinal pattern analysis, introduced by Bandt and Pompe (2002), provides a robust framework for studying time series without requiring distributional assumptions. This approach transforms successive time-window segments of a time series into symbols according to the ordinal relationships among the observations. These symbols, known as ordinal patterns, form an empirical probability distribution that is invariant under monotonic transformations and is resilient to noise and outliers.

Information theoretic descriptors computed from ordinal pattern distributions provide a compact summary of the underlying dynamics. The permutation entropy quantifies the unpredictability of the ordinal pattern distribution. Statistical complexity combines entropy with disequilibrium to characterize the balance between disorder and structure. Complementary to these global descriptors, the Fisher Information Measure (FIM) captures local variations in the ordinal pattern distribution and is sensitive to subtle structural changes. Together, these descriptors provide a rich characterization of temporal dynamics.

II. Methodology

Our working hypothesis is that the temporal dynamics of different Land Use/Land Cover (LULC) classes produce distinct levels of randomness and structure in SAR time series data. We expect these differences to be reflected in

ordinal pattern distributions constructed using an embedding dimension of $D = 5$. Furthermore, we hypothesize that the resulting distributions can be characterized by Shannon entropy, statistical complexity, Fisher Information Measure, and Fisher complexity, enabling discrimination among the LULC classes.

A. Ordinal Patterns

Let $\mathbf{x} = (x_1, x_2, \dots, x_{n+D-1})$ be a real valued time series, where $D \geq 2$ is the user-defined embedding dimension and n is the number of overlapping windows. For each $t = 1, 2, \dots, n$, define the window

$$s_t = (x_t, x_{t+1}, \dots, x_{t+D-1}).$$

Let $\pi^1, \pi^2, \dots, \pi^k$ denote the $k = D!$ possible ordinal patterns associated with embedding dimension D. For each window s_t, the corresponding ordinal pattern is the permutation that orders its components in increasing order. This produces the symbolic sequence

$$\pi = (\pi_1, \pi_2, \dots, \pi_n), \qquad \pi_t \in \{\pi^1, \pi^2, \dots, \pi^k\}.$$

Let $N_\ell = \#\{t : \pi_t = \pi^\ell\}$, $\ell = 1, 2, \dots, k$, be the number of times the ordinal pattern π^ℓ occurs in π. The empirical ordinal pattern probability function is

$$\hat{\mathbf{p}} = \left(\frac{N_1}{n}, \frac{N_2}{n}, \dots, \frac{N_k}{n}\right) = (\hat{p}_1, \hat{p}_2, \dots, \hat{p}_k).$$

B. Permutation Entropy

The normalized Shannon entropy of $\hat{\mathbf{p}}$, called Permutation entropy, is

$$H(\hat{\mathbf{p}}) = -\frac{1}{\log k} \sum_{\ell=1}^{k} \hat{p}_\ell \log \hat{p}_\ell.$$

This is a quantitative measure of the uncertainty encoded in the ordinal pattern distribution: low values indicate strong regularity or determinism, while high values reflect randomness or unpredictability. The normalization ensures that $0 \leq H(\hat{\mathbf{p}}) \leq 1$, allowing comparisons across different embedding dimensions.

Entropy alone cannot distinguish between randomness and structured complexity, motivating the introduction of complementary measures.

C. Fisher Information Measure

As an additional descriptor, we consider the Fisher Information Measure. Following Olivares et al. (2012), FIM captures local variations in the ordinal pattern distribution and is sensitive to structural changes that may not be reflected by entropy alone. The discrete form adopted in this study is given by Ferri et al. (2009):

$$F(\hat{\mathbf{p}}) = \frac{1}{4} \sum_{\ell=1}^{k-1} \frac{2(\hat{p}_{\ell+1} - \hat{p}_\ell)^2}{(\hat{p}_{\ell+1} + \hat{p}_\ell)}.$$

D. Disequilibrium

Permutation entropy and FIM provide complementary information about the ordinal pattern distribution. However, neither descriptor quantifies the degree to which the distribution deviates from the uniform distribution. To incorporate this information, we consider the disequilibrium introduced by Lopez-Ruiz et al. (1995) with the uniform distribution $\mathbf{u} = (1/k, 1/k, \dots, 1/k)$ as reference.

Let $S(\mathbf{p}) = -\sum_{\ell=1}^{k} p_\ell \log p_\ell$ denote the unnormalized Shannon entropy, so that $H(\mathbf{p}) = S(\mathbf{p})/\log k$. The Jensen–Shannon divergence between $\hat{\mathbf{p}}$ and $\mathbf{u}$ is

$$J(\hat{\mathbf{p}}, \mathbf{u}) = S\left(\frac{\hat{\mathbf{p}} + \mathbf{u}}{2}\right) - \frac{1}{2} S(\hat{\mathbf{p}}) - \frac{1}{2} \log(k).$$

It is a widely adopted, bounded, and information-theoretically meaningful disequilibrium measure. Its maximum possible value is

$$J_{\max} = -\frac{1}{2}\left[\frac{(k+1)}{k} \log(k+1) - 2\log(2k) + \log k\right].$$

Using this, the normalized disequilibrium is defined as

$$Q(\hat{\mathbf{p}}) = \frac{J(\hat{\mathbf{p}}, \mathbf{u})}{J_{\max}}.$$

This quantity satisfies $Q(\hat{\mathbf{p}}) = 0$ for the uniform distribution, and $Q(\hat{\mathbf{p}}) = 1$ when all probability mass is concentrated on a single pattern.

E. Shannon and Fisher Complexities

Following Lamberti et al. (2004), statistical complexity combines an information measure and a disequilibrium. In this study, we consider two complexity measures.

$$C_S(\hat{\mathbf{p}}) = H(\hat{\mathbf{p}}) Q(\hat{\mathbf{p}}),$$

and

$$C_F(\hat{\mathbf{p}}) = F(\hat{\mathbf{p}}) Q(\hat{\mathbf{p}}).$$

F. Data Representation

The Sentinel-1 SAR dataset consists of temporal cross-polarized (VH: Vertical Transmit, Horizontal Receive) backscatter observations acquired over Haldwani, Uttarakhand, India, located in the sub-Himalayan region and encompassing the Terai Central Forest Division, a managed forest area, as shown in Fig.~1. Single Look Complex (SLC) data were downloaded from the Alaska Satellite Facility Distributed Active Archive Center (ASF-DAAC) for the period 2017–2023.

The data were pre-processed using the InSAR Scientific Computing Environment (ISCE3) to generate gamma-nought backscatter products, with NASADEM at 1-arcsecond resolution used for terrain correction (Rosen et al., 2018). We consider $T \geq 2$ images acquired at approximately 12-day temporal intervals. These images

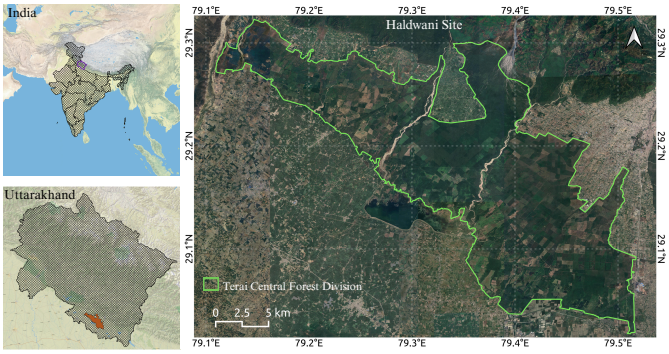


Figure 1: Study area map showing Haldwani in the Uttarakhand State in India. The green polygon outlines the Terai Central Forest Division.

were resampled to a spatial resolution of 25 m and co-registered using PLAnT (Polarimetric-interferometric Lab and Analysis Tools) to the common spatial support

$$S = \{1, 2, \dots, R\} \times \{1, 2, \dots, C\},$$

where each image contains R rows and C columns (Lavalle et al., 2016).

Field visits were conducted in 2023 and 2025 to collect geolocated reference information for different LULC classes. This information was combined with photo-interpretation of temporal high-resolution optical imagery to identify representative samples belonging to five classes: Agriculture, Deforested, Forest, Urban, and Water. For each class, forty homogeneous regions covering a 2×2 pixel neighbourhood were selected for subsequent analysis.

The SAR image data consist of temporal backscatter observations recorded at each spatial location. We have $T \geq 2$ images co-registered to the same support. Let $z(i, j, t)$ denote the real-valued measurement at position $(i, j) \in S$ and time $1 \leq t \leq T$. The original time series at site $(i, j) \in S$ is denoted by

$$\mathbf{z}(i, j) = (z(i, j, 1), z(i, j, 2), \dots, z(i, j, T)).$$

We augment the original time series with surrounding measurements from a 2×2 neighbourhood of each position, omitting edge and corner pixels. Such augmentation can be organised in two ways: Stacked, where the four measurements observed at time t are contiguous in the augmented time series, and Interleaved, where time-consecutive measurements are contiguous. The stacked time series is defined as

$$\begin{aligned}\mathbf{z}^{\mathrm{S}}(i, j) = \Big(& z(i, j, 1), z(i+1, j, 1), z(i+1, j+1, 1), z(i, j+1, 1), \\ & z(i, j, 2), z(i+1, j, 2), z(i+1, j+1, 2), z(i, j+1, 2), \\ & \dots, \\ & z(i, j, T), z(i+1, j, T), z(i+1, j+1, T), \\ & z(i, j+1, T) \Big).\end{aligned}$$

Thus, observations from neighbouring pixels are grouped together at each time instant, producing a sequence of length $4T$.

The interleaved time series is defined as

$$\begin{aligned}\mathbf{z}^{\mathrm{I}}(i, j) = \Big(& z(i, j, 1), z(i, j, 2), \dots, z(i, j, T), \\ & z(i+1, j, 1), z(i+1, j, 2), \dots, z(i+1, j, T), \\ & z(i+1, j+1, 1), z(i+1, j+1, 2), \dots, \\ & z(i+1, j+1, T), \\ & z(i, j+1, 1), z(i, j+1, 2), \dots, z(i, j+1, T) \Big).\end{aligned}$$

This representation also produces a sequence of length $4T$, but preserves the temporal ordering within each neighbouring pixel before moving to the next spatial location.

Preliminary experiments were conducted using the Original, Stacked, and Interleaved representations. The Stacked representation was selected for the subsequent analysis because it incorporates spatial neighbourhood information while maintaining temporal correspondence among neighbouring observations at each acquisition time.

To illustrate the effect of the data transformation procedure, three time series representations—Original, Interleaved, and Stacked—are shown in Fig.~2. The example time series are selected from the same SAR dataset and are presented only to demonstrate the construction of the different representations. The subsequent analysis is performed using the LULC classes described previously.

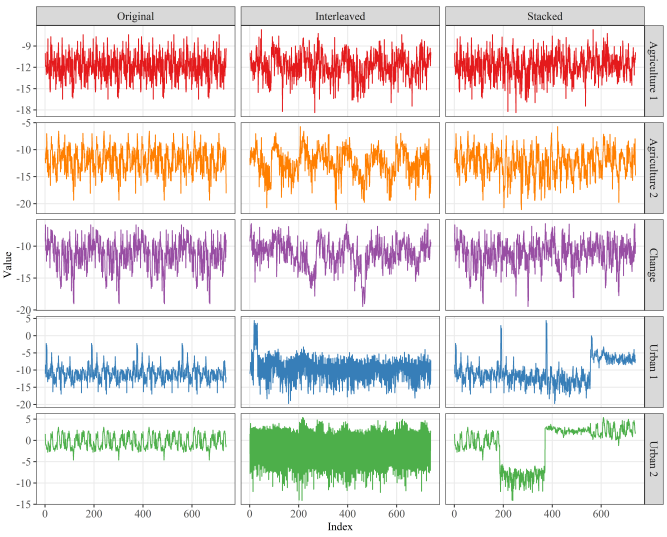


Figure 2: Time series for all pixel types across data structures.

G. Proposed Methodology

The proposed framework begins with the extraction of SAR time series from the five LULC classes and their

transformation into the Stacked representation using 2×2 pixel neighbourhoods. Ordinal pattern distributions are then computed using embedding dimension $D = 5$, from which four descriptors are obtained: permutation entropy H, Shannon statistical complexity C_S, Fisher Information Measure F, and Fisher complexity C_F. The resulting features are analysed using the $H \times C_S$ and $F \times C_F$ planes and are subsequently used as inputs to an SVM classifier for LULC classification. For classification, the samples are divided into training and testing sets using stratified sampling, and the SVM hyperparameters are selected by cross-validation on the training data.

III. Results

Fig.~3 presents the resulting Shannon and Fisher information-complexity planes for the stacked representation with embedding dimension $D = 5$. The Shannon plane shows substantial overlap among Forest, Deforested, Urban, and Water samples, while Agriculture exhibits comparatively greater separation. Similar behaviour is observed in the Fisher plane, where Agriculture and, to a lesser extent, Water display improved visual separation from the remaining classes.

These results indicate that the ordinal pattern features capture meaningful temporal characteristics of the SAR time series. However, the substantial overlap among several LULC classes suggests that visual inspection of the information-complexity planes alone is insufficient to assess class separability. A supervised classification approach was therefore employed to evaluate the discriminative ability of the extracted features.

The discriminative ability of the four ordinal pattern features was evaluated using a SVM with an RBF kernel. A stratified 70 %–30 % training-testing split was used. The SVM hyperparameters were selected through a grid-search procedure using the training data with 10-fold cross-validation: $C \in \{0.1, 1, 10, 100, 1000\}$ and $\gamma \in \{0.01, 0.05, 0.1, 0.5, 1, 5\}$. The minimum cross-validation error was obtained with $C = 100$ and $\gamma = 0.1$, and the resulting model was evaluated on the independent test set.

Classification performance was assessed using sensitivity and balanced accuracy. Sensitivity quantifies the proportion of correctly identified samples within each class, while balanced accuracy accounts for both sensitivity and specificity. Fig.~4 provides a two-dimensional visualization of the four ordinal pattern features using the first two principal components, which together explain 99.12 % of the total variance. PCA was used only for visualization, while the SVM was trained using the four original ordinal pattern features. The projection reveals partial separation among some LULC classes, while considerable overlap remains, particularly among Urban, Water, and Deforested samples.

The classification performance reported in Table~I is consistent with the feature overlap observed in Fig.~4.

Table I: SVM classification performance metrics

Class	Sensitivity	Balanced Accuracy
Agri	0.833	0.885
Deforested	0.500	0.677
Forest	0.917	0.844
Urban	0.167	0.552
Water	0.333	0.635

Agriculture and Forest achieved the highest sensitivities of 83.3 % and 91.7 %, with balanced accuracies of 88.5 % and 84.4 %, respectively. Deforested showed intermediate performance, with 50.0 % sensitivity and 67.7 % balanced accuracy. Urban and Water were the most difficult classes to distinguish, with sensitivities of 16.7 % and 33.3 %, and balanced accuracies of 55.2 % and 63.5 %, respectively.

Overall, the results indicate that ordinal-pattern-based features provide a useful representation of SAR temporal dynamics, with greater separability observed for LULC classes exhibiting distinct temporal behaviour and reduced separability for classes with similar temporal characteristics.

IV. Conclusion

The results indicate that ordinal-pattern-based features provide a useful representation of SAR temporal dynamics for LULC characterization. The information-theoretic descriptors capture complementary aspects of ordinal pattern distributions, enabling discrimination among classes with distinct temporal behaviour while highlighting similarities among classes with comparable temporal characteristics. The stacked representation incorporates information from neighbouring spatial locations while maintaining temporal correspondence among observations.

The proposed methodology has minimal computational requirements, making it suitable for large-scale SAR time series analysis. By efficiently characterizing temporal behaviour, ordinal-pattern-based features provide a promising framework for future studies of SAR time series and their underlying temporal dynamics.

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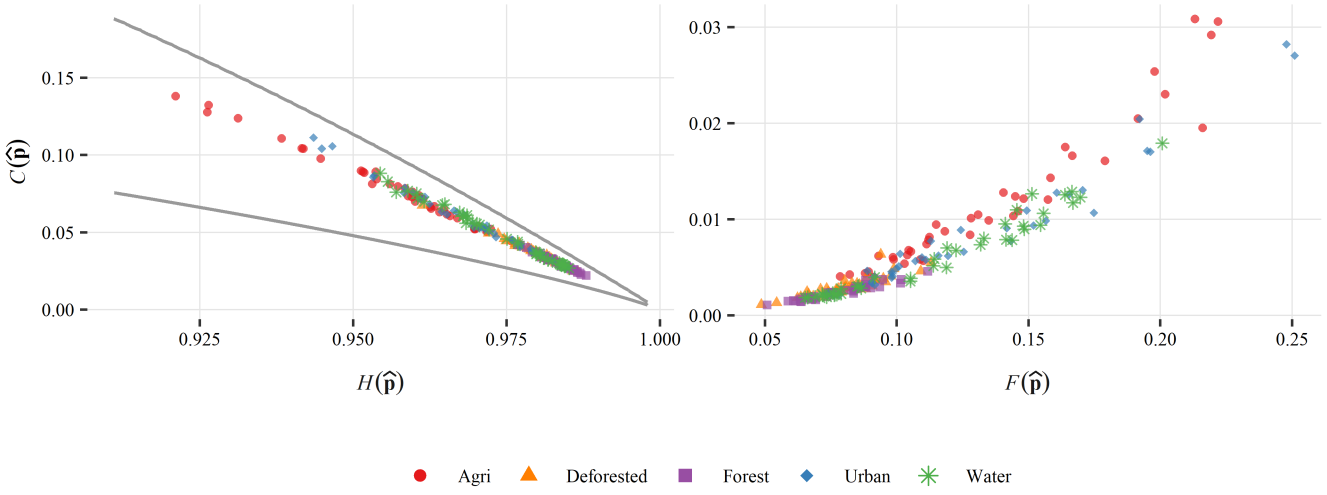


Figure 3: Shannon entropy-complexity and Fisher information-complexity representations for the five land cover classes.

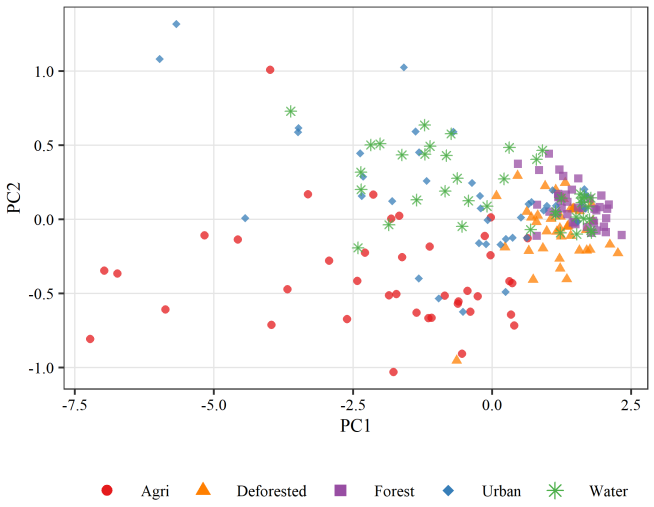


Figure 4: Projection of the four ordinal pattern features onto the first two principal components for the five land cover classes.

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